Kinematics and integration

1 The velocity $v\,(t)$, in metres per second, of an object travelling horizontally along a straight line after t seconds is modelled by:

$$v\,(t) = 12t,\ \text{where } t \geq 0$$

The object is initially at the origin.

a Find the displacement $x\,(t) = \int v(t)\,dt$ of the particle at time t.

b Find the time at which $x(t) = 54\ \text{m}$.

2 The velocity $v\,(t)$, in metres per second, of an object travelling horizontally along a straight line after t seconds is modelled by:

$$v\,(t) = 6t + 10,\ \text{where } t \geq 0$$

The object is initially $8\ \text{m}$ to the right of the origin.

a Find the displacement $s\,(t)$ of the particle at time t.

b Find the displacement of the object after 5 seconds.

3 The velocity $v\,(t)$, in metres per second, of an object travelling horizontally along a straight line after t seconds is modelled by:

$$v\,(t) = 12t^2 + 30t + 9,\ \text{where } t \geq 0$$

The object is initially $6\ \text{m}$ to the left of the origin.

a Find the displacement, $s\,(t)$, of the particle at time t.

b Find the displacement of the object after 5 seconds.

4 The velocity $v\,(t)$, in metres per second, of an object travelling horizontally along a straight line after t seconds is modelled by:

$$v\,(t) = 12\sqrt{t}$$

The object is initially $5\ \text{m}$ to the right of the origin.

a Find the displacement, $x\,(t)$, of the particle at time t.

b Hence, find the position of the object after 9 seconds.

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5 The velocity $v\,(t)$, in metres per second, o[redacted]rticle travelling horizontally along a straight line after t seconds is given by:

$$v\,(t) = 30\sqrt{t}$$

a Find the velocity of the particle after 4 seconds.

b Sketch a graph of the velocity of the particle over the first 10 seconds.

c Find the area under the velocity function from $t = 0$ to $t = 4$.

d Describe what the area found in part (c) represents.

6 The velocity $v\,(t)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\,(t) = 2 - \frac{t}{3}$$

a Find the velocity of the particle after 9 seconds.

b Sketch a graph of the velocity of the particle over the first 12 seconds.

c Find the area bounded between the velocity function and the horizontal axis from $t = 0$ to $t = 9$.

d Describe what the area found in part (c) represents.

7 The velocity $v\,(t)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\,(t) = 6t + 15$$

The displacement after 2 seconds is 45 metres to the right of the origin.

a Calculate the initial velocity of the particle.

b Find the displacement, $x\,(t)$, of the particle at time t.

c Calculate the displacement of the particle after four seconds.

d Find the total distance travelled between two and four seconds.

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8 The velocity $v(t)$, in metres per second, o[redacted]rticle travelling horizontally along a straight line after t seconds is given by:

$$v(t) = 4t - 4$$

The particle is instantaneously stationary when it is $1\ \mathrm{m}$ right of the origin.

a Find the time when the particle is stationary.

b Find the displacement, $x(t)$, of the particle at time t.

c Find the value of $x(t)$ at:

i $t = 0$ ii $t = 1$ iii $t = 2$

9 The velocity $v(t)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v(t) = 6t^2 - 14t + 3$$

The displacement after 2 seconds is 4 metres to the left of the origin.

a Calculate the initial velocity of the particle.

b Find the displacement, $x(t)$, of the particle at time t.

c Calculate the displacement of the particle after four seconds.

d Assuming the object continues to move in the same direction, find the total distance travelled between two and four seconds.

10 The velocity $v(t)$, in metres per second, of an object travelling horizontally along a straight line after t seconds is given by:

$$v(t) = 12t^2 - 48t, \text{ where } t \geq 0$$

The object is initially $5\ \mathrm{m}$ to the right of the origin.

a Find the displacement, $x(t)$, of the particle at time t.

b Find the times, t, when the object is at rest.

c Find the displacement at which the object is stationary other than its initial position.

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11 The velocity $v\left(t\right)$, in metres per second, o[redacted]rticle travelling horizontally along a straight line after t seconds is given by:

$$v\left(t\right)=3t^{\frac{3}{2}}-2t$$

a Find the initial acceleration of the particle.

b If the particle is initially at the origin, find an expression for the displacement $x(t)$.

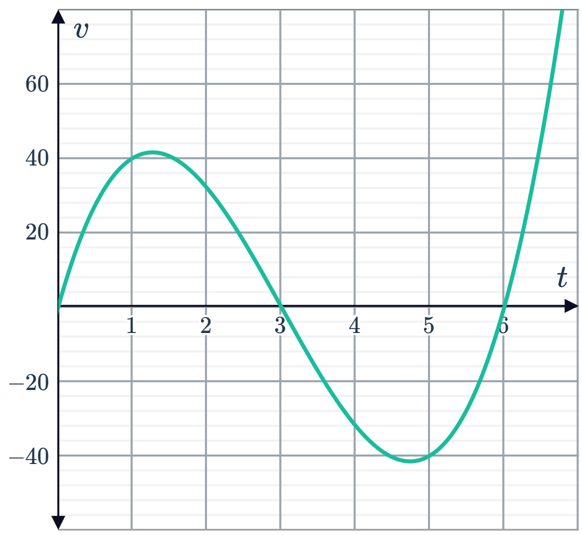
c Find the exact time $t>0$ at which the particle is stationary.

d How far does the particle travel in the first 2 seconds? Round your answer correct to two decimal places.

12 A pen moves along the x axis ruling a line.
The diagram shows the graph of the velocity of the tip of the pen as a function of time.
The velocity, in centimetres per second, after t seconds is given by:

$$v\left(t\right)=4t^{3}-36t^{2}+72t$$

When $t=0$, the tip of the pen is at $x=5$.



a Find an expression for $x\left(t\right)$, the position of the tip of the pen, as a function of time.

b What feature will the graph of $x\left(t\right)$ have at the point where $t=3$ and how does this relate to the movement of the pen?

c The pen uses 0.02 milligrams of ink per centimetre travelled. Find the amount of ink used between $t=0$ and $t=4$.

13 The velocity $v\left(t\right)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\left(t\right)=5\left(1-e^{-t}\right)$$

a Calculate the exact total change in displacement over the interval $[0,5]$.

b Hence, calculate the average change in displacement over the interval $[0,5]$, correct to three decimal places.

c Use technology to calculate the average change in the distance over the interval $[0,5]$, correct to three decimal places.

d For the given velocity function is the average change in displacement equal to the average change in distance for $t\geq 0$? Explain why or why not.

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14 The acceleration $a\,(t)$, in $\mathrm{m/s^2}$, of an objec[redacted]elling horizontally along a straight line after t seconds is modelled by:

$$a = 2t - 15, \text{where } t \geq 0$$

The object is initially moving to the right at $56\ \mathrm{m/s}$.

a Find the velocity, $v\,(t)$, of the particle at time t.

b Find for all times at which the particle is at rest.

15 The acceleration $a\,(t)$, in $\mathrm{m/s^2}$, of an object travelling horizontally along a straight line after t seconds is modelled by:

$$a\,(t) = 6t - 27,\ \text{where } t \geq 0$$

After 10 seconds, the object is moving at $90\ \mathrm{m/s}$ in the positive direction.

a Find the velocity, $v\,(t)$, of the particle at time t.

b Find all the times at which the particle is at rest.

16 The acceleration $a\,(t)$, in $\mathrm{m/s^2}$, of an object travelling horizontally along a straight line after t seconds is modelled by:

$$a\,(t) = 2,\ \text{where } t \geq 0$$

The object is initially $8\ \mathrm{m}$ to the right of the origin and moving to the left at $5\ \mathrm{m/s}$.

a Find the velocity, $v\,(t)$, of the particle at time t.

b Find the displacement, $x\,(t)$, of the particle at time t.

c Find the position of the object at 7 seconds.

d Find the time at which the particle is moving at a speed of $3\ \mathrm{m/s}$ to the right.

17 A coin is thrown vertically upwards from the top of a building $30\ \mathrm{m}$ high with an initial velocity of $20\ \mathrm{m/s}$. Using the approximation of acceleration due to gravity of $-10\ \mathrm{m/s^2}$, find:

a The time, t, taken for the coin to reach its maximum height.

b The maximum height reached by the coin.

c The time, t, taken for the coin to reach the ground.

d The velocity of the coin as it hits the ground.

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18 An object is thrown vertically upwards with an initial velocity of $15 \mathrm{~m/s}$ and an initial height of $2 \mathrm{~m}$. Using the approximation of acceleration due to gravity of $-9.8 \mathrm{~m/s^2}$, find:

a The time, t, taken for the object to reach its maximum height.

b The maximum height reached by the object, correct to one decimal place.

c The time, t, taken for the object to reach the ground, correct to one decimal place.

d The velocity of the object as it hits the ground, correct to one decimal place.

19 The acceleration $a(t)$, in $\mathrm{cm/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = -e^{4t}$$

The particle is initially at rest.

a Find the velocity, $v(t)$, of the particle at time t.

b Find the change in the particle's position after 5 seconds, to the nearest centimetre.

20 The acceleration $a(t)$, in $\mathrm{m/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = e^{-t}$$

The particle starts off from the origin with an initial velocity of $2 \mathrm{~m/s}$.

a Find the velocity of the particle after 4 seconds, correct to two decimal places.

b Find the displacement of the particle after 4 seconds, correct to the nearest hundredth of a metre.

21 The acceleration $a(t)$, in $\mathrm{m/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = 2e^{3t}$$

The initial velocity of a particle is $10 \mathrm{~m/s}$.

a Find its velocity, $v(t)$, after $t = 2$ seconds. Round your answer to the nearest metre per second.

b Find the displacement of the particle after 4 seconds, given the initial position is $2 \mathrm{~m}$ to the left of the origin. Round your answer to the nearest metre.

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Trigonometric applications

22 The velocity $v\left(t\right)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\left(t\right)=6\sin 3t$$

a Find the acceleration of the particle at $t=\frac{\pi}{3}$.

b Find the displacement of the particle at $t=\frac{\pi}{9}$, given that when $t=0$, the displacement is 2 m.

23 The velocity $v\left(t\right)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\left(t\right)=6\cos 3t$$

a Find the acceleration of the particle at $t=\frac{\pi}{9}$.

b Find the displacement of the particle at $t=\frac{\pi}{3}$, given that when $t=0$, the displacement is 3 m.

24 The velocity $v\left(t\right)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\left(t\right)=4\sin\left(\frac{\pi t}{3}\right)$$

a Find the change in displacement in the first 8 seconds.

b Given that the distance function is the absolute value of the displacement function, calculate the change in distance in the first 4 seconds.

25 The velocity $v\left(t\right)$, in metres per second, of a particle travelling horizontally along a straight line after t seconds is given by:

$$v\left(t\right)=6\cos\left(\frac{\pi t}{4}\right)$$

a Find the change in displacement in the first 9 seconds.

b Find the change in distance in the first 8 seconds.

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26 The acceleration $a(t)$, in $\mathrm{m/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = 64 \sin 4t$$

a Find $v(t)$, the velocity in $\mathrm{m/s}$ at time t, if the particle is initially traveling at $2\ \mathrm{m/s}$.

b Hence, calculate the speed of the particle at $t = \frac{\pi}{3}$.

c Find $x(t)$, the displacement in metres at time t, if the particle is initially located $3\ \mathrm{m}$ from the origin.

d Hence, calculate the displacement at $t = \frac{\pi}{3}$.

27 The acceleration $a(t)$, in $\mathrm{m/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = 50 \cos 5t$$

a Find $v(t)$, the velocity in $\mathrm{m/s}$ at time t, if the particle is initially traveling at $6\ \mathrm{m/s}$.

b Hence, calculate the speed of the particle at $t = \frac{\pi}{6}$.

c Find $x(t)$, the displacement in metres at time t, if the particle is initially located at $3\ \mathrm{m}$ from the origin.

d Hence, calculate the displacement of the particle at $t = \frac{\pi}{6}$.

28 The acceleration $a(t)$, in $\mathrm{m/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = 6 \sin\left(\frac{\pi t}{3}\right)$$

a Find the change in velocity between $t = 1$ and $t = 7$.

b Given that the speed function is the absolute value of the velocity function, calculate the total change in speed in the initial 4 seconds.

29 The acceleration $a(t)$, in $\mathrm{m/s^2}$, of a particle travelling horizontally along a straight line after t seconds is modelled by:

$$a(t) = 4 \cos\left(\frac{\pi t}{4}\right)$$

a Find the change in velocity between $t = 1$ and $t = 3$.

b Calculate the total change in speed in the first 4 seconds.

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